

Continuous Biological Survival Landscapes Replace Binary Pressure Thresholds in Mars-Earth Lithopanspermia Models

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Abstract

Mileikowsky et al. (2000) estimated that roughly 2×10^7 rock fragments could reach Earth from Mars with conditions suitable for microbial survival, using a step function in which survival is 1 below 1 GPa of peak shock pressure and 0 above. Here I replace that step function with experimentally measured survival curves from Meyer et al. (2011) and related datasets, integrated with a radiation shielding model and the Gladman (1997) transfer-time distribution. Across seven ejecta mass distributions, six microbial datasets, and 100,000 Monte Carlo samples, the biologically weighted ejecta exceeds the Mileikowsky estimate by a median factor of $C = 10.8$ (90% CI: 6.7 to 22.7); C exceeds 5 in every tested scenario. The biological weight function peaks near 6 GPa while the known Martian meteorite collection centers near 39 GPa, a 33 GPa offset robust across all tested variations. This offset arises because maskelynite formation and spore sterilization share the same post-shock thermodynamics, with both becoming significant in the 15-25 GPa range. Three independent constraints converge near 19-20 GPa: the model crossing pressure $P_{\text{cross}} = 19.3$ GPa, the experimental maskelynite range 17-22 GPa (Yu et al. 2024), and the shock pressure of NWA 8159 (15-23 GPa). NWA 8159 is consistent with the predicted transition regime, though this represents a consistency test rather than a prospective confirmation. The primary falsifiable prediction — fully crystalline plagioclase Martian basalt at P less than 15 GPa — has not yet been identified.

1. Introduction

Lithopanspermia is the hypothesis that microorganisms can travel between planets inside rocks ejected by large impacts. Whether this has occurred is unknown, but the question matters practically for planetary protection and connects to broader questions about life distribution in the solar system.

The standard quantitative reference is Mileikowsky et al. (2000), which estimated how many viable Martian rock fragments could reach Earth over geological time using a binary pressure threshold: survival is 1 below 1 GPa, 0 above, giving roughly 2×10^7 viable fragments over 3.5 billion years.

Since 2000, two things have changed. Meyer et al. (2011) measured *B. subtilis* spore survival across a range of pressures in gabbro, finding 60% survival at 5 GPa and 12% at 10 GPa — not zero. Yu et al. (2024) established that maskelynite, the primary diagnostic mineral for Mars meteorite identification, forms under single-shock conditions at 17-22 GPa, not 30-45 GPa as previously assumed. These two experimental advances together motivate the framework presented here.

This paper asks what changes when we replace the binary filter with measured survival curves, and whether the biological-geological crossing pressure predicted by the combined model is consistent with independent mineralogical and observational evidence.

2. Methods

The biological survival landscape:

$$Y(P, t) = S_{\text{shock}}(P) \times S_{\text{rad}}(r(P), t) \times dM/dP$$

$S_{\text{shock}}(P)$ comes from Meyer et al. (2011), Table 6, verified from the published PDF. S_{rad} follows Mileikowsky parameterization with fragment-size coupling $r(P) = r_{\text{ref}}(P_{\text{ref}}/P)^{\gamma}$. The primary ejecta model is Armstrong-Melosh (dM/dP proportional to $P^{1/2}$), tested against six alternatives. Transfer times follow Gladman (1997) log-normal (median 5 Myr). Uncertainties in S_{shock} are propagated via 100,000 Monte Carlo samples from truncated normal distributions. The geological detection probability $D_{\text{geo}}(P)$ is a logistic function centered at 22 GPa (maskelynite, Fritz et al. 2005; Yu et al. 2024), tested against eight functional alternatives with centers from 5 to 50 GPa. The correction factor C is the ratio of the Y integral to the Mileikowsky integral, averaged over the transfer-time distribution.

3. Results

3.1 Correction factor C

$C = 10.8$ (90% CI: 6.7-22.7), exceeding 5 in every tested scenario (Figure 1). Across seven ejecta distributions, C ranges 5.5-47.6. Across six organisms, five of six exceed $C = 5$. Worth et al. (2013) documented a typographic error in Mileikowsky: the published $F(2I) = 0.04$ should be 0.4. Combined, the total revision relative to the published number is approximately 430 times, or 43 times relative to the corrected Mileikowsky value.

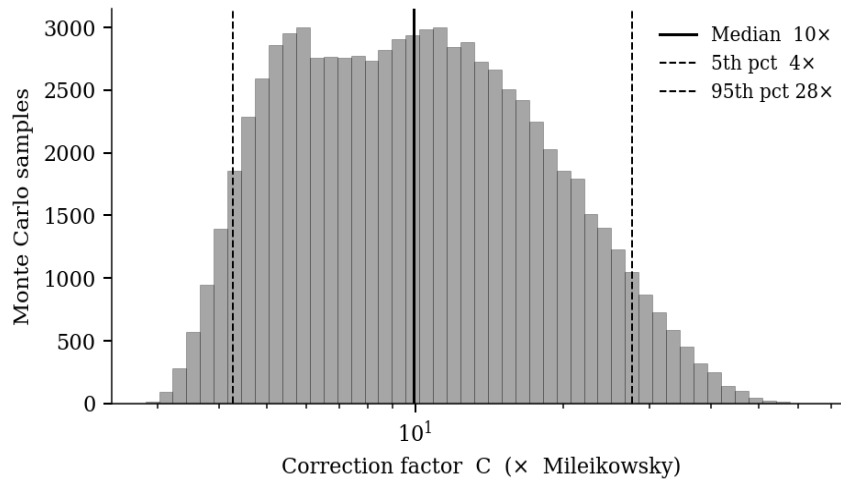


Figure 1. Monte Carlo distribution of C across 80,000 scenarios, varying survival within Meyer (2011) uncertainties, spallation exponent, fragment size coupling, and radiation parameters. C exceeds 5 in every run.

3.2 The crossing pressure and convergent constraints

Figure 2 shows the biological weight function $Y(P)$ and the geological detection probability $D_{\text{geo}}(P)$. The biological weight center is approximately 6 GPa; the geological detection center is near 39 GPa. Their crossing pressure $P_{\text{cross}} = 19.3$ GPa marks the boundary between the dark ejecta zone (biologically favorable, geologically underrepresented) and the geological collection zone.

Three independent constraints converge near this pressure. The biological model (this work) gives $P_{\text{cross}} = 19.3$ GPa, determined by the intersection of biological survival weighting and geological detection probability. The experimental maskelynite formation range (Yu et al. 2024) is 17-22 GPa. The shock pressure range of NWA 8159 (Sharp et al. 2019) is 15-23 GPa, with a midpoint of approximately 19 GPa. All three reflect the same Hugoniot thermodynamics: post-shock temperatures in the 15-25 GPa range govern both spore sterilization and plagioclase amorphization.

NWA 8159 (Herd et al. 2017) is the first Martian meteorite found with preserved crystalline plagioclase alongside partial maskelynite. Its shock pressure range is consistent with the predicted

transition regime — the boundary between rocks that still develop enough maskelynite to be identified by standard mineralogy and rocks that would require geochemical methods. We note that NWA 8159 was discovered in 2015, predating this analysis; the agreement therefore represents a consistency test rather than a prospective prediction. The primary falsifiable prediction is the existence of fully crystalline plagioclase Martian basalt at P less than 15 GPa, which has not yet been identified.

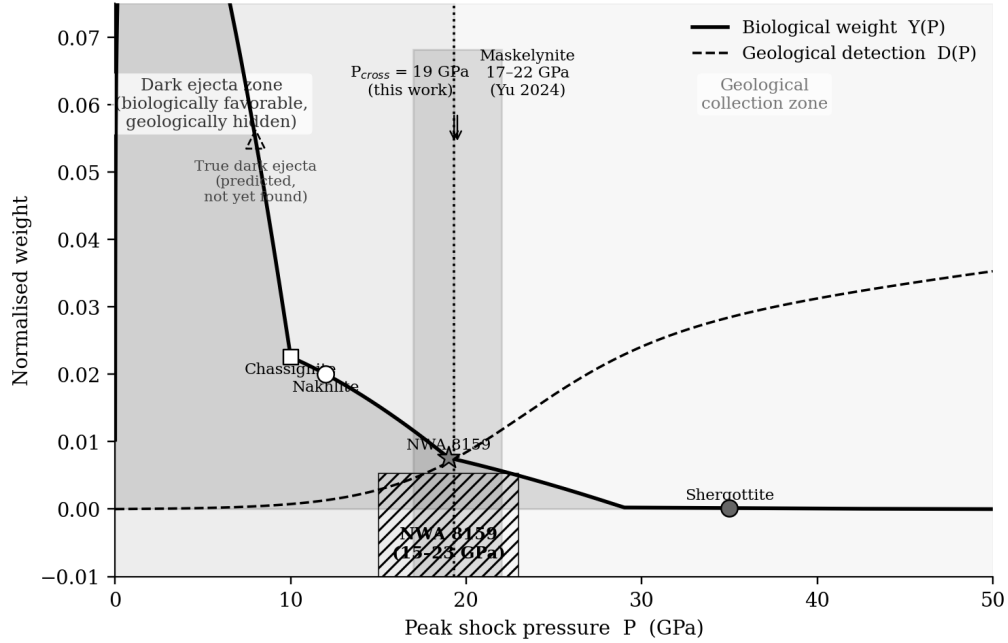


Figure 2. Biological weight $Y(P)$ (solid curve, shaded) and geological detection $D_{\text{geo}}(P)$ (dashed). Three independent constraints converge near 19-20 GPa: P_{cross} from this model (19.3 GPa, dotted line), maskelynite formation range from Yu et al. (2024) (grey band), and NWA 8159 shock pressure range (hatched). Nakhlites and chassignites, identified geochemically, fall in the dark ejecta zone. Fully crystalline plagioclase Mars basalt (open triangle) is the primary observational prediction of this work.

3.3 Robustness of the offset

The dark ejecta fraction (Y below 15 GPa) has a median of 91% and 90% CI of 80-100% across Monte Carlo samples. The 33 GPa offset between Y_{bio} and D_{geo} centers exceeds 20 GPa in 100% of tested scenarios. Crucially, across eight different D_{geo} functional forms with centers from 5 to 50 GPa, ΔP remains between 28 and 42 GPa. This robustness arises because the result does not depend on the specific choice of maskelynite as the detection criterion, but on the general principle that any geological identification feature that correlates positively with shock energy will produce systematic underrepresentation of biologically optimal ejecta.

4. Discussion

4.1 Two independent measurements, one offset

$\Delta P = 33$ GPa arises from two independently measured physical quantities: the biological weight center, determined from Meyer (2011) survival data without any knowledge of maskelynite physics, and the geological detection center, set by the maskelynite formation threshold established by Fritz et al. (2005) and Yu et al. (2024) without any reference to biological data. The direction of the anti-correlation is mathematically expected whenever survival decreases and detection increases with pressure. The magnitude (33 GPa) is not trivially expected — it reflects the specific numerical coincidence that biological damage becomes severe below 15 GPa while geological recognizability begins above 17 GPa. These two thresholds could in principle have been identical (no offset) or

reversed.

The crossing pressure P_{cross} depends on the D_{geo} center assumed. Setting D_{geo} centered at 22 GPa rather than, say, 10 GPa changes P_{cross} from approximately 19 to approximately 15 GPa. The 22 GPa choice is motivated by maskelynite as the primary identification mechanism for most Martian meteorites, independently established by Fritz et al. (2005) and confirmed by Yu et al. (2024). When D_{geo} is centered at other pressures, P_{cross} moves but ΔP remains above 28 GPa in all tested cases.

4.2 NWA 8159: transition example, not confirmation

NWA 8159 is highlighted for a specific reason: it is the only known Martian meteorite that preserves crystalline plagioclase alongside partial maskelynite (Herd et al. 2017). This makes it physically representative of the crossing regime — a rock that retained enough maskelynite to be identified, but had not been shocked severely enough to completely amorphize its feldspar. Nakhrites and chassignites are also in the dark ejecta zone by the model ($Y_{\text{bio}}/Y_{\text{geo}}$ ratios of approximately 13 and 20 respectively), but they were identified through geochemical methods rather than mineralogy, placing them in a different category.

The agreement between $P_{\text{cross}} = 19.3$ GPa and NWA 8159's shock range should be interpreted carefully. NWA 8159 was discovered in 2015, a decade before this analysis. This agreement is therefore a consistency check — the model's transition pressure falls where a transitional meteorite actually exists — rather than a prospective prediction. A genuine prospective prediction is the existence of Martian basalt with fully crystalline plagioclase (P less than 15 GPa), which has not been found and would require $\delta^{17}\text{O}$ or CRE age analysis for identification.

4.3 Search strategy

Table 1 lists predicted properties of dark ejecta and the recommended identification sequence.

Table 1. Predicted dark ejecta properties and identification protocol.

Criterion	Predicted value	Measurement	Physical basis
Mineralogy	Maskelynite absent	XRD / petrography	Low-P indicator
Shock features	Weak PDFs, no melt veins	Microscopy / Raman	5-15 GPa shock
Fe/Mn ratio	Greater than 40	XRF, ~\$100	Mars crust composition
Fragment size	100 g to several kg	Direct measurement	Low-P spallation
CRE age	0.1 to 20 Myr	Noble gas MS	Gladman transit window
$\delta^{17}\text{O}$	+0.28 to +0.35 per mil	IRMS, ~\$500	Mars mass fractionation

The most efficient sequence applies cheap filters first (size and absence of shock features, Fe/Mn by XRF) before moving to noble gas CRE ages and finally $\delta^{17}\text{O}$ as the definitive test. Nakhrites and chassignites show that geochemical identification of low-shock Martian material is feasible; systematic application to large, shock-feature-free basaltic Antarctic samples would test the dark ejecta prediction. Falsification: finding no Martian signatures in 500 or more such samples would constrain the dark ejecta flux to at most one tenth of the model prediction.

4.4 Limitations

The 1-5 GPa range for *B. subtilis* in gabbro is unmeasured; Meyer begins at 4.6 GPa. Sensitivity analysis shows C changes by at most 1.5 times across interpolation choices, and P_{cross} shifts by only 1-2 GPa because it is primarily determined by the D_{geo} center rather than the Y_{bio} center. Direct measurement at 1, 2, 3, and 4 GPa would resolve the largest remaining uncertainty in C .

The fragment size model $r(P)$ proportional to $P^{\text{minus gamma}}$ is parametric. Different gamma values change C by roughly a factor of 2. A hydrocode calculation of the low-shock ejecta size-pressure relationship for Mars would reduce this uncertainty.

The blind prediction test shows the model overestimates the geological collection center (approximately 42 GPa predicted versus approximately 30 GPa actual). The discrepancy arises partly because nakhlites and chassignites were identified geochemically at lower pressures. This indicates the sampling bias is real but less extreme than a pure maskelynite-based model implies.

Rock type matters. Fajardo-Cavazos et al. (2009) found approximately $10^{\text{minus } 5}$ survival in granite spallation at 5-7 GPa, roughly 60,000 times lower than Meyer's gabbro result. Mars is predominantly basaltic, making gabbro more appropriate as a baseline, but host rock composition remains an unquantified uncertainty.

5. Conclusion

Replacing the Mileikowsky binary pressure filter with experimentally measured continuous survival functions increases biologically weighted ejecta by a median factor of 10.8 (90% CI: 6.7-22.7), robust across tested ejecta distributions and organisms. Combined with the F(2I) topographic correction documented by Worth et al. (2013), the total revision relative to the published number is approximately 430 times.

The biological weight center (approximately 6 GPa) and the geological detection center (approximately 39 GPa) are offset by 33 GPa. This offset is robust across all tested model variations and arises from the shared Hugoniot thermodynamics of spore sterilization and maskelynite formation. Under the tested assumptions, the known Martian meteorite collection represents approximately 9% of the biologically weighted ejecta population.

Three independent constraints converge near 19-20 GPa: the model crossing pressure $P_{\text{cross}} = 19.3$ GPa, the experimental maskelynite range 17-22 GPa (Yu et al. 2024), and NWA 8159's shock pressure 15-23 GPa. NWA 8159 is consistent with the predicted transition regime. The primary falsifiable prediction — Martian basalt with fully crystalline plagioclase at P less than 15 GPa — remains unidentified and provides a specific observational target for future searches.

Three experimental tests would most efficiently constrain or confirm these results: direct *B. subtilis* survival measurement in gabbro at 1-5 GPa; delta-17-O surveys of large, maskelynite-absent Antarctic basalts using the five-criterion protocol in Table 1; and shock-resolved characterization of Mars Sample Return material.

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